

Yingrui Hou

Simulation & Validation Engineer | Physical Modelling, Test Correlation & Technical Application

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SUMMARY

Simulation and validation engineer with a PhD in experimental physics and several years of experience building C++/Python workflows for calibration, validation, inference, and monitoring on complex physical measurement systems.

My core strength is making imperfect models useful for engineering decisions by connecting simulation, measurements, and physics-based interpretation through simulation-to-test alignment, bias correction, dataset preparation, and uncertainty-aware analysis.

My BSc in Applied Physics included solid mechanics and finite element analysis coursework, and I ramp quickly on new physical domains. Combined with a PhD in experimental physics, this gives me a strong base for tyre, chassis, and vehicle-system simulation topics.

CORE SKILLS

- Simulation and modelling: physical-system modelling, simulation-to-measurement alignment, Geant4, ROOT / RooFit
- Validation and test correlation: calibration, bias correction, anomaly diagnostics, uncertainty-aware analysis
- Dataset and model preparation: response-map reconstruction, model correction, statistical modelling, monitoring
- Engineering workflow: Python, C/C++, Linux, reproducible analysis pipelines, shared codebases
- Technical collaboration and foundation: multidisciplinary coordination, technical communication, solid mechanics and FEA coursework

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Built reusable C++/Python workflows that connected simulation, noisy measurements, and physics-based interpretation for decision-facing analyses on complex systems.
- Combined test-beam data, simulation, and response maps to quantify how non-ideal behaviour affected overall performance, including a quantified contribution below 1% in a detailed study.
- Developed calibration and correction workflows to reduce structured bias and improve model credibility across operating conditions.
- Built monitoring and diagnostic pipelines that traced discrepancies back to detector behaviour or software issues and supported concrete corrective actions.
- Worked in shared Linux-based codebases with reproducible workflows, collaborative review discipline, and frequent interaction with multidisciplinary experts.

SELECTED OUTCOMES

- Quantified that response non-uniformity contributed less than 1% to system-resolution degradation in a detailed measurement-to-model study.
- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis and calibration on realistic measurement data.

Education: Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021 | BSc in Applied Physics - China University of Mining and Technology, 2010-2014 | Languages: Chinese: Native speaker, English: C1, French: B1 to B2